

Alpha-Beta Divergence For Variational Inference

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Problem: Standard VI with KL-divergence has limitations

- ▶ Mode-seeking behavior (underestimates variance)
- ▶ Sensitive to outliers in data

Existing solutions address only one issue:

- ▶ Rényi divergence: controls mass-covering
- ▶ Gamma divergence: controls robustness

Goal: Unified framework for both properties

Variational Inference Recap

Posterior approximation: $p(\theta|X) \approx q(\theta)$

Standard objective: minimize $D[q(\theta)||p(\theta|X)]$

ELBO formulation:

$$\mathcal{L}_D(q, X, \varphi) = \log p(X|\varphi) - D(q(\theta)||p(\theta|X, \varphi))$$

Challenge: Choice of divergence D affects approximation quality

- ▶ Mass-covering vs mode-seeking
- ▶ Robustness to outliers
- ▶ Computational tractability

Problems with KL Divergence

Formula:

$$D_{\text{KL}}(q\|p) = \int q(\theta) \log \frac{q(\theta)}{p(\theta)} d\theta$$

Gradient:

$$\frac{d}{d\varphi} D_{\text{KL}}(q\|p) = - \int \frac{dq(\theta)}{d\varphi} \left[\log \frac{p(\theta)}{q(\theta)} - 1 \right] d\theta$$

Key issue: Log-term increases cost of overestimating p

Result:

- ▶ Underestimation of posterior variance (Turner & Sahani, 2011)
- ▶ Mode-seeking behavior: focuses on single mode
- ▶ Not robust to outliers

Renyi and Gamma divergences

$$D_{\alpha}(q\|p) = \frac{1}{\alpha - 1} \log \int q(\theta)^{\alpha} p(\theta)^{1-\alpha} d\theta$$

$$\begin{aligned} D_{\gamma}^{\beta}(q\|p) &= \frac{1}{\beta(\beta + 1)} \log \int p(\theta)^{\beta+1} d\theta \\ &\quad - \frac{1}{\beta} \log \int p(\theta) q(\theta)^{\beta} d\theta \\ &\quad + \frac{1}{\beta + 1} \log \int q(\theta)^{1+\beta} d\theta \end{aligned}$$

Scale Invariant Alpha-Beta (sAB) Divergence

Formula:

$$\begin{aligned} D_{\text{sAB}}^{\alpha, \beta}(p \| q) = & \frac{1}{\beta(\alpha + \beta)} \log \int p(\theta)^{\alpha + \beta} d\theta \\ & + \frac{1}{\alpha(\alpha + \beta)} \log \int q(\theta)^{\alpha + \beta} d\theta \\ & - \frac{1}{\alpha\beta} \log \int p(\theta)^{\alpha} q(\theta)^{\beta} d\theta \end{aligned}$$

Two control parameters: α and β

- ▶ Extended to entire $\mathbb{R}^2$ domain
- ▶ Smooth interpolation between divergences

Special Cases of sAB Divergence

Divergence	Parameters
KL divergence	$(\alpha, \beta) = (0, 1)$
Reverse KL	$(\alpha, \beta) = (1, 0)$
Rényi divergence	$\alpha + \beta = 1$
Gamma divergence	$\alpha = 1$
Hellinger distance	$(\alpha, \beta) = (0.5, 0.5)$
Chi-square divergence	$(\alpha, \beta) = (2, -1)$

Control Parameters: α and β

Use $\lambda = \alpha + \beta$ and β for interpretation

λ (**robustness parameter**):

- ▶ $\lambda < 2$: robust to outliers
- ▶ $\lambda > 2$: focuses on outliers

β (**mass-covering parameter**):

- ▶ $\beta > 1$: mass-covering behavior
- ▶ $\beta < 1$: mode-seeking behavior

sAB Divergence Gradient

$$\frac{d}{d\varphi} D_{\text{sAB}}^{\alpha,\beta}(q\|p) = -\frac{1}{\beta} \left[\frac{\int \frac{dq(\theta)}{d\varphi} q(\theta)^{\alpha+\beta-1} \left(\frac{p(\theta)}{q(\theta)} \right)^{\beta} d\theta}{\int q(\theta)^{\alpha} p(\theta)^{\beta} d\theta} - \frac{\alpha}{\beta} \frac{\int \frac{dq(\theta)}{d\varphi} q(\theta)^{\alpha+\beta-1} d\theta}{\int q(\theta)^{\alpha+\beta} d\theta} \right]$$

Key terms:

- ▶ $q(\theta)^{\alpha+\beta-1}$: controls robustness
- ▶ $\left(\frac{p(\theta)}{q(\theta)} \right)^{\beta}$: controls mass-covering

sAB Variational Objective

Direct divergence optimization (not ELBO!)

$$\begin{aligned} D_{\text{sAB}}^{\alpha, \beta}(q(\theta) \| p(\theta | X)) &= \frac{1}{\alpha(\alpha + \beta)} \log \mathbb{E}_q \left[\frac{p(\theta, X)^{\alpha + \beta}}{q(\theta)^{\alpha + \beta}} \right] \\ &\quad + \frac{1}{\beta(\alpha + \beta)} \log \mathbb{E}_q \left[q(\theta)^{\alpha + \beta - 1} \right] \\ &\quad - \frac{1}{\alpha\beta} \log \mathbb{E}_q \left[q(\theta)^{\alpha + \beta - 1} \left(\frac{p(\theta, X)}{q(\theta)} \right)^\beta \right] \end{aligned}$$

- ▶ All terms are expectations w.r.t. q
- ▶ $p(X)$ terms cancel out
- ▶ Monte Carlo approximation with K samples

Monte Carlo approximation:

- ▶ Sample $\theta_1, \dots, \theta_K \sim q(\theta)$
- ▶ Approximate each expectation term
- ▶ Use $K \approx 100$ samples

Gradient computation:

- ▶ Reparameterization trick for differentiability
- ▶ Can leverage modern VI optimization methods
- ▶ ADAM optimizer with learning rate 0.01

Experimental Setup

Two main experiments:

1. **Synthetic data:** Bayesian linear regression with 5% corrupted data
2. **UCI datasets:** Bayesian neural networks with artificial outliers

Network architecture:

- ▶ 2-layer with 50 hidden units
- ▶ ReLU activation
- ▶ Fully factorized Gaussian approximation

Parameter selection:

- ▶ Cross-validation to select optimal (α, β)
- ▶ Compare with KL, Rényi, and Gamma divergences

Results: Synthetic Data

Bayesian linear regression with 5% corrupted data

Method	(λ, β)	MAE	MSE
KL	(1.0, 0.0)	0.58 ± 0.001	0.53 ± 0.003
Rényi	(1.0, 0.3)	0.58 ± 0.003	0.51 ± 0.007
Gamma	(1.8, 0.8)	0.34 ± 0.025	0.21 ± 0.030
sAB	(1.9, -0.3)	0.34 ± 0.025	0.21 ± 0.030

Conclusion: sAB and Gamma both ignore outliers effectively, while standard VI is highly sensitive to corruption

Bayesian neural networks with 10% outliers

Dataset	KL	sAB
Boston Housing	1.13 ± 0.043	1.07 ± 0.016 (1.25, -0.5)
Concrete	1.16 ± 0.002	1.07 ± 0.008 (1.5, -0.25)
Yacht	1.09 ± 0.025	1.05 ± 0.011 (1.25, -0.25)

Conclusion: sAB outperforms KL-VI when outliers present

Key Insights

1. **Unified framework:** mass-covering + robustness in one objective
2. **Direct divergence optimization** (not ELBO)
3. **Smooth interpolation** between known divergences
4. **Two interpretable parameters** with clear effects:
 - ▶ λ controls outlier sensitivity
 - ▶ β controls mode-seeking vs mass-covering
5. **Cross-validation** can find optimal parameters
6. **Most effective** when $\lambda < 2$ and $\beta < 1$